SOFT NON-DIAGONALITY PENALTY ENABLES LATENT SPACE-LEVEL INTERPRETABILITY OF PLM AT NO PERFORMANCE COST

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ABSTRACT

Emergence of large scale protein language models (pLMs) has led to significant performance gains in predictive protein modeling. However, it comes at a high price of interpretability, and efforts to push representation learning towards explainable feature spaces remain scarce. The prevailing use of domain-agnostic and sparse encodings in such models fosters a perception that developing both parameter-efficient and generalizable models in a low-data regime is not feasible. In this work, we explore an alternative approach to develop compact models with interpretable embeddings while maintaining competitive performance. With the BiLSTM-AE model as an example trained on positional property matrices, we introduce a soft weight matrix non-diagonality penalty. Through Jacobian analysis, we show that this penalty aligns embeddings with the initial feature space while leading to a marginal decrease in performance on a suite of four common peptide biological activity classification benchmarks. Moreover, it was demonstrated that the use of one-hot encoded sequence clustering-based contrastive loss to produce semantically meaningful latent space allows to further improve benchmarking performance. The use of amino acid physicochemical properties and DFT-derived cofactor interaction energies as input features provides a foundation for intrinsic interpretability, which we demonstrate on fundamental peptide properties. The resulting model is over 33,000 times more compact than the state-of-the-art pLM ProtT5. It demonstrates performance stability across diverse benchmarks without task-specific fine-tuning, showcasing that domain-tailored architectural design can yield highly parameter-efficient models with fast inference and preserved generalization capabilities.

1 INTRODUCTION

Machine learning (ML) achieved significant success in the field of drug discovery (Jorner et al., 2021; Lee et al., 2020), due to SMILES representations (Weininger, 1988), which can be transformed into molecular graphs. Such graphs are used to calculate interpretable chemical descriptors using RDKit (Landrum et al., 2013) or Mordred (Moriwaki et al., 2018). However, their application becomes challenging for biopolymers due to their higher structural complexity. Being a class of biopolymers, peptides are sequences of $<$100 amino acids. Due to the much larger contact surface offering numerous hydrogen bonds and side-chain contacts, peptides are highly effective and selective agents for diagnostics and therapy (Henninot et al., 2018; Fisher et al., 2019; Peterson & Barry, 2018; Torres et al., 2019). Their spatial organization is governed by multiple factors, including hydrogen bonding, hydrophobic interactions, coordination, disulfide bridges etc. (Gregorc et al., 2023; Rezai et al., 2006; Victorio & Sawyer, 2023). Furthermore, their activity depends also on contextual information about neighboring monomers hard to capture using descriptors designed for molecular graphs. In peptide research, ML methods are also commonly employed (Basith et al., 2020), with two major approaches used for their representation, which are property- and sequence-based (Xu et al., 2020). While the former approach utilizes interpretable peptide physicochemical properties and is unable to consider sequence context, the latter is contextual but lacks chemistry and interpretability. Moreover, majority of the models based on these approaches are highly task-specific, limiting their potential use. To address these challenges, it is beneficial to combine property- and sequence-based approaches into a unified framework.

In this work, we explore an alternative approach to develop compact models with interpretable embeddings while maintaining competitive performance. With the BiLSTM-AE model as an example trained on positional property matrices, we introduce a soft weight matrix non-diagonality penalty. Through Jacobian analysis, we show that this penalty aligns embeddings with the initial feature space while leading to even marginal increase in performance on a suite of four peptide biological activity classification benchmarks. Moreover, it was demonstrated that the use of one-hot encoded sequence clustering-based contrastive loss to produce semantically meaningful latent space allows to further improve benchmarking performance. The use of amino acid physicochemical properties and DFT-derived cofactor interaction energies as input features provides a foundation for intrinsic interpretability, which we demonstrate on easy computable fundamental peptide properties for clarity of relationships. For more fair evaluation of our model embeddings on benchmark datasets, average Shannon entropies and Levenshtein distances were introduced to consider peptide sequence complexity an roughly estimate overall dataset complexity, since machine learning (ML) models' performance on classification benchmark datasets using conventional encoding strategies and embeddings was observed not to be directly determined by the number of samples alone. Being interpretable and >4,000, >12,000, and >6,000 times more compact than ProtBERT (Elnaggar et al., 2022), Ankh-large (Elnaggar et al., 2023), and ESM-C (600M) (ESM Team, 2024), respectively, our model demonstrates competitive results. It outperforms all these baselines on the anti-inflammatory peptides (AIP) prediction task, matches the performance of Ankh-large on antimicrobial peptides (AMP), is comparable with ESM-C on anti-oxidative peptides (AOP), and bypasses ProtBERT on AOP and anti-diabetic peptides (ADP) predictions. Based on the Matthew's correlation coefficient (MCC), our approach also shows the second-best average performance stability across all tasks after Ankh-large. The resulting model is over >33,000 times more compact than the state-of-the-art (SOTA) protein language model (pLM) ProtT5 (3B) (Elnaggar et al., 2022), demonstrating performance stability across diverse benchmarks without task-specific fine-tuning, showcasing that domain-tailored architectural design can yield highly parameter-efficient models with fast inference and preserved generalization capabilities. Although we successfully show how to develop extremely compact, generalizable, and feature space-level interpretable pLM compatible with top-performance large scale models, we found scalability of such models is not straightforward and requires more research to compete or surpass the current SOTA ProtT5 raw performance.

Key contributions:

- soft penalty to off-diagonal weight matrix elements of BiLSTM-AE architecture coupled with forced diagonal weight initialization was introduced to provide feature disentanglement and align the latent space with the initial amino acid physico-chemical properties represented as positional property matrices, thereby introducing latent space-level interpretability
- latent space-level interpretability was demonstrated via feature importance and correlation analysis on total of four regression tasks related to fundamental physico-chemical properties of peptides
- being interpretable, the final model is comparable or even surpasses the most common ProtBERT, ankh, and ESM-C pLMs in fair benchmarking on a suite of four peptide biological activity classification tasks
- it was found that performance of the vast majority of current pLM embeddings is still comparable to conventional peptide one-hot encoding strategy, which highlights the persistence of representation learning quality problem in pLMs and the questionable rationale behind the development of domain-agnostic large scale models pre-trained on large corpora of randomly picked sequences
- non-diagonality loss term providing latent space-level interpretability was shown to marginally compromise benchmarking performance given the main architecture and training setup proposed in this work
- the final model was shown to be 3-4 orders more parameter-efficient in comparison with the most common ProtBERT, ESM-C, Ankh, and current SOTA ProtT5 in raw performance (>4,000, >6.000, >12,000, and >33,000, respectively) being trained on at least 2 orders smaller datasets without loss of generalization, where no fine-tuning was performed before benchmarking

2 Related works

Current SOTA pLMs are predominantly based on the Transformer architecture (Vaswani et al., 2017), pre-trained on massive datasets of protein sequences. Below, we review key representatives, focusing on their architectural choices and, crucially, their approaches to interpretability.

2.1 ProtBERT and ProtT5

ProtBERT and ProtT5 adapt the canonical BERT (Devlin et al., 2019) and T5 (Raffel et al., 2020) architectures to the protein domain. Their key strength lies in effective transfer learning, achieving strong performance on downstream tasks like secondary structure prediction and remote homology detection. The primary method of interpretation for these models is the analysis of self-attention maps, which aim to reveal residue-residue interactions important for the model's predictions. However, this approach has significant limitations: attention weights are known to be noisy, difficult to aggregate across layers and heads, and do not necessarily equate to importance or causal relationships (Jain & Wallace, 2019). Thus, while providing some insight, they offer a post-hoc and often ambiguous interpretation of the model's internal reasoning.

2.2 ESM

The Evolutionary Scale Modeling (ESM) family, including the 600M-parameter ESM-Cambrian, scales Transformer-based pLMs to unprecedented sizes. The main advantage of these models is their capacity to capture complex evolutionary patterns and achieve SOTA performance on zero-shot fitness prediction and structure prediction tasks. Similar to ProtBERT/T5, their interpretability relies heavily on analyzing attention heads to identify functional sites or structural contacts. While some heads can learn biologically meaningful patterns, the interpretation remains indirect and is not an inherent property of the learned representations themselves.

2.3 Ankh

Ankh presents a comprehensive empirical study of architectural choices for protein language modeling. Through extensive ablation studies, the authors identify optimal configurations for span masking, activation functions and relative positional embeddings. The key contribution is the systematic exploration of design space, resulting in two models with optimized encoder-decoder architecture. While achieving competitive performance, the model follows the standard Transformer paradigm and consequently inherits its interpretability limitations. Similar to other pLMs, understanding model decisions relies on post-hoc analysis of attention mechanisms rather than intrinsic interpretability built into the representation learning process.

3 Methods

3.1 Context and motivation

The method proposed in the paper constructs property matrices for peptides from physico-chemical descriptors of 20 canonical residues and 2 prevalent modifications, using 43 molecular and 3 DFT-derived features. While convolutional autoencoders (CAE) provided the baseline, recurrent and transformer variants were introduced to capture long-range context. Embeddings are further refined using InfoNCE-based contrastive loss: k-means clustering, specifically *MiniBatchKMeans* from *sklearn.cluster* module, of one-hot encoded peptides supplies pseudo-labels for positive/negative pairs, aligning the interpretable property space with empirically effective one-hot representation. To force the best model to align embedding features softly with the initial physico-chemical properties, non-diagonality penalty was applied to matrix weights, where per-feature relationships were quantified using Jacobian matrix (see 3.7 for details).

3.2 Data collection

For model training, unlabeled unique peptide sequences containing no extra monomers were retrieved from the NCBI database (Sayers et al., 2023), yielding 6,749,334 sequences. A sampling procedure was implemented to reduce data volume and computational costs while preserving diversity, grouping peptides by the most frequently occurring amino acids to maintain balanced composition over sequence length distribution. Consequently, small (**S**) and big (**B**) datasets were constructed through clustering peptide sequences based on amino acid frequency vectors, with the number of clusters equal to the unique amino acids. Then the number of unique clusters was identified based on the most represented amino acid for stratified sampling, and the minimum cluster size was determined to prevent under-representation of smaller clusters. A small fraction (1%) of sequences from each cluster formed dataset **S**; a medium fraction (10%) was sampled for dataset **B**. Finally, the subsets from all clusters were combined to create the final datasets. Clustering-based stratified sampling facilitated the formation of manageable datasets, where the final model was trained using the dataset **S** comprising 155,920 sequences. Final data scaling experiment was performed on dataset **B** comprising 467,792 sequences.

3.3 Molecular descriptors

A total of 43 molecular RDKit descriptors were used to characterize the properties of amino acids (see **Appendix A**), which were normalized to a range of [-1, 1]. Based on the descriptor matrix and DFT descriptors (see **Section 3.4** for details), feature arrays were constructed for each sequence with padding applied to align sequences to a uniform length of 96 amino acids.

3.4 DFT-derived descriptors

This group of descriptors includes three parameters derived from the interaction energies of amino acids with divalent metal cations: calcium (Ca^{2+}), magnesium (Mg^{2+}), and barium (Ba^{2+}), obtained through density functional theory (DFT) calculations. Detailed methodology is described in Hu et al. (2022), with data available in the NOMAD database (Draxl & Scheffler, 2019). The study modeled 23,243 conformers of amino acids and their complexes with these cations, identifying 21,909 stationary points on potential energy surfaces with energy ranges up to 4 eV (390 kJ/mol). Free energy values were extracted using the NOMAD API and categorized into four groups: complexes with Mg^{2+}, Ca^{2+}, and Ba^{2+} cations, and isolated amino acid conformers. Free energy values for isolated cations were also retrieved from NOMAD. For each amino acid, average free energy values across all conformers were calculated, and interaction descriptors were derived by normalizing the differences between complex and isolated free energy values.

3.5 AE-based architectures

Convolutional an Variational Autoencoders (CAE/VAEs). Initially, CAE-based models with per-feature convolutions were tested as a baseline to study the model performance when features are not mixed at all due to architectural restrictions. Then, we implemented a probabilistic framework by replacing the deterministic latent space of CAE with a variational latent space. This approach encourages smoother, more continuous latent representations, potentially mitigating sparsity. We experimented with multiple VAE variants, including β-VAE and InfoVAE adjusting hyperparameters e.g., KL divergence weight (β) to balance reconstruction accuracy and latent space regularization.

BiLSTM. Given the sequential nature of data, we employed BiLSTM-based architectures well-suited for capturing long-range dependencies. Unlike unidirectional recurrent models, BiLSTM leverages both forward and backward contexts, enabling a more comprehensive understanding of sequence information. We hypothesized that this bidirectional approach would enhance the quality of the generated embeddings, making them more effective for downstream tasks.

Transformer-Based Models. Motivated by the success of transformer architectures in natural language processing, we implemented a transformer-based AE. Transformers leverage self-attention mechanisms to capture global dependencies across sequences, making them particularly promising for peptide data.

Each of the architectures described in this section was trained using the same dataset **S** and evaluated on identical downstream tasks to ensure a fair comparison with the baseline CAE (see **Appendix B**).

3.6 CONTRASTIVE LEARNING IMPLEMENTATION

Since all the models performed comparably or worse than conventional sequence encodings, contrastive learning was implemented, which strengthens embeddings by drawing positive pairs together in latent space while separating negatives.

Positive and Negative Pair Construction. Given the unique properties of peptide sequences, we designed a domain-specific strategy for constructing positive and negative pairs:

- Positive Pairs: Positive pairs were obtained by *MiniBatchKMeans* clustering of one-hot encoded peptides; k was selected *via* the Elbow criterion, and peptides sharing a cluster were deemed positives.
- Negative Pairs: Peptides from different clusters were treated as negatives.

Contrastive Loss Function. We utilized contrastive loss based on the Information Noise-Contrastive Estimation (InfoNCE) objective:

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\frac{\text{sim}(\mathbf{z},\mathbf{z}^+)}{\tau})}{\sum_{j=1}^{K} \exp(\frac{\text{sim}(\mathbf{z},\mathbf{z}_j)}{\tau})}, \tag{1}$$

where $\mathbf{z} = E(\mathbf{x})$, $E(\cdot)$ is the encoder, $\text{sim}(\cdot,\cdot)$ denotes a similarity measure (e.g., cosine similarity), K is the number of negative samples in a batch, and $\tau > 0$ is a temperature hyperparameter.

3.7 SOFT NON-DIAGONALITY PENALTY

To address the poor performance of convolutional autoencoders on our benchmark tasks, we adopted a Bidirectional LSTM (BiLSTM) architecture. However, the BiLSTM's inherent tendency for feature entanglement necessitated the explicit induction of feature disentanglement during training.

To achieve this, we employed two complementary techniques. First, we applied a soft penalty specifically to the off-diagonal elements of the BiLSTM weight matrices. For a weight matrix $M \in \mathbb{R}^{n\times k}$, the penalty term is defined as the mean of squared off-diagonal elements:

$$\mathcal{L}_M = \frac{\|M \odot (\mathbf{1} - I)\|_F^2}{n \cdot k} \tag{2}$$

where $\odot$ denotes the Hadamard product, I is the identity matrix, and $\mathbf{1}$ is a matrix of ones. The total penalty is the sum over all weight matrices, scaled by a coefficient λ:

$$\mathcal{L}_{\text{off-diag}} = \lambda \sum_{M\in\mathcal{W}} \mathcal{L}_M \tag{3}$$

Here, $\mathcal{W}$ comprises all *input-to-hidden* and *hidden-to-hidden* weight matrices of the BiLSTM layers. Second, we used forced diagonal weight initialization. The combined effect encourages the model to maintain nearly diagonal weight transformations throughout training.

To quantitatively evaluate the degree of feature disentanglement, we propose a diagonality metric based on the encoder's Jacobian. We compute the mean absolute Jacobian $\bar{J} \in \mathbb{R}^{D_{in}\times D_{out}}$ over the dataset. The diagonality metric is then defined as:

$$\mathcal{D} = \frac{\sum_i \bar{J}_{ii}}{\sum_{i,j} \bar{J}_{ij}} \tag{4}$$

A value of $\mathcal{D}$ close to 1 indicates a highly diagonal transformation and thus successful feature disentanglement.

3.8 COMPUTATIONAL RESOURCES

All calculations were performed on a server with a general configuration consisting of 6 A6000 GPUs, 256 cores, AMD EPYC 7763 64-Core Processor, 512 GB RAM. The training procedure was performed using 2 GPUs and 50 GB RAM.

4 EXPERIMENTS

4.1 BENCHMARK DATASETS

We performed benchmarking of all the models mentioned in the paper on four public peptide biological activity classification datasets: antimicrobial (AMP) (Cao et al., 2023), anti-inflammatory (AIP) (Raza et al., 2023), antidiabetic (ADP) (Chen et al., 2022), and antioxidant (AOP) (Qin et al., 2023). Dataset statistics covering size, sequence complexity, and data imbalance are summarized in **Table 1**. Sequence complexity is described here by several parameters including length statistics, as well as average Shannon entropy characterizing the extent of non-equiprobability of amino acids per sequence and average Levenshtein distance showing inter-sequence dissimilarity. From the analysis, it follows that ADP can be considered as the hardest dataset based on dataset size and high values of average Shannon entropy, Levenshtein distance, as well as length spread.

Table 1: Main statistics of benchmark datasets. Average Levenshtein distance characterizes the diversity of sequences by measuring the pairwise differences. Average Shannon entropy characterizes the diversity of amino acids within a single sequence by quantifying the average level of uncertainty associated with all types of amino acids.

Dataset	Num. of peptides	Avg. Levenshtein distance	Avg. Shannon entropy	Avg. length	Min. length	Max. length	Pos/Neg sample ratio
ADP	472	19.20 ± 17.39	3.22 ± 0.16	19.60 ± 25.86	11	41	1.00
AIP	3790	15.48 ± 5.52	3.15 ± 0.11	16.39 ± 6.86	11	30	0.76
AMP	8268	18.38 ± 17.70	3.03 ± 0.26	18.53 ± 28.52	11	30	1.00
AOP	2120	7.12 ± 12.83	2.03 ± 0.48	5.92 ± 13.37	2	20	1.00

4.2 EXPERIMENTAL SETUP

This section presents the experimental framework for evaluating the quality and generalizability of the embeddings generated by our model. The assessment was conducted on four peptide classification datasets encompassing distinct biological activities (detailed in Section 4.1). To isolate the evaluation on the embedding quality itself, rather than the complexity of a deep learning classifier, we employed a classical gradient boosting model as a simple yet effective downstream predictor (Shwartz-Ziv & Armon, 2022). The primary metric for comparison was the Matthews Correlation Coefficient (MCC), chosen for its informativeness and robustness to class imbalance (Chicco & Jurman, 2020) — a property particularly relevant for datasets like AIP, as evidenced by the statistics in Table 1. To ensure a precise and reliable estimate of performance, we employed a rigorous 5-fold cross-validation protocol, reporting both the mean metric values and their standard errors (the latter indicated in parentheses).

Our comparative analysis includes three conventional peptide encoding methods: one-hot encoding, BLOSUM62, and 3-mer counts, and several SOTA pLMs: ProtBERT, Ankh-large, ESM-C (600M), and ProtT5. The following subsections detail the development and optimization experiments for our model, with results provided in **Appendixes C** and **F** (Tables 6 through 14). The final comparative benchmarking results against all baselines are consolidated in Section 4.5 (**Table 2**).

4.3 AE-BASED ARCHITECTURES BENCHMARKING

The first experiment involves CAE architecture with per-feature convolutions and feature space size equal to the number of physico-chemical features trained on positional property matrices, which we treat as a baseline, where no feature mixing occurs due to architectural restrictions. Results for all encodings with the CAE model are shown in **Table 6**; additional metrics also can be found in **Appendix C**.

To study which architecture fits the objective best, we screened total of 80 models from four AE-based groups: CAEs, VAEs (incl. Info-VAE, β-VAE), BiLSTM-AEs, and transformer-AEs, with each trained under diverse hyperparameters (see **Appendix B**). Model embeddings were evaluated with the same downstream model as the baseline and benchmark encodings (see **Section 4.2**). **Table 7** summarizes the results, where BiLSTM-AE was chosen for further optimization.

4.4 Clustering-based contrastive learning

Introducing a supervised contrastive loss (**Section 3.6**) into the BiLSTM-AE — motivated by the strong one-hot baseline (**Table 6**) — clustered similar peptides and separated dissimilar ones, markedly boosting MCC and all other metrics. Contrastive BiLSTM-AE (cBiLSTM-AE) results versus benchmark encodings appear in **Table 2**.

4.5 Soft non-diagonality penalty

To enforce feature disentanglement within the model, we applied a soft penalty to the off-diagonal elements of the weight matrices (as detailed in Section 3.7). Our evaluation comprises two main parts: (1) a comprehensive benchmarking of cBiLSTM-AE model with soft non-diagonality penalty (dcBiLSTM-AE) against all baselines (see section 4.2 for details), and (2) a scalability analysis where the model was trained on a large-scale dataset **B**. The results of both the comparative benchmarking and the scalability experiment are consolidated in **Table 2**.

Table 2: dcBiLSTM-AE model benchmarking. No preliminary fine-tuning on tasks was performed for any of the models.

Encoding type	Model size	Dataset for training	**MCC (5-fold cross validation)**				Avg.	Min	Max
			Anti diabetic	Anti inflammatory	Anti microbial	Anti oxidant			
One-hot	N/A	N/A	0.197 (0.015)	0.216 (0.015)	0.560 (0.004)	0.764 (0.013)	0.434	0.197	0.764
Blosum	N/A	N/A	0.026 (0.020)	0.290 (0.012)	0.337 (0.015)	0.189 (0.012)	0.211	0.026	0.337
Threemers	N/A	N/A	0.131 (0.060)	**0.357 (0.009)**	0.519 (0.003)	0.539 (0.015)	0.387	0.131	0.539
ProtBert	420M	217M	0.334 (0.031)	0.138 (0.021)	0.658 (0.007)	0.580 (0.011)	0.428	0.138	0.658
ESM C	600M	-	0.433 0.017	0.193 (0.012)	**0.679 (0.005)**	0.761 (0.019)	0.517	0.193	0.761
Ankh	1.15B	59M	**0.574) (0.032)**	0.335 (0.011)	0.614 (0.007)	**0.863 (0.015)**	**0.596**	0.335	**0.863**
BiLSTM-AE	1.3M	**0.15M**	-0.004 (0.053)	0.342 (0.009)	0.415 (0.005)	0.496 (0.031)	0.312	-0.004	0.496
cBiLSTM-AE	1.3M	**0.15M**	0.277 (0.065)	0.316 (0.009)	0.569 (0.002)	0.692 (0.012)	0.464	0.277	0.692
dcBiLSTM-AE	**90K**	**0.15M**	0.371 (0.039)	0.355 (0.004)	0.611 (0.006)	0.775 (0.018)	0.528	**0.355**	0.775
dcBiLSTM-AE	**90K**	0.47M	0.327 (0.042)	0.337 (0.009)	0.599 (0.009)	0.750 (0.016)	0.503	0.327	0.750
ProtT5	3B	45M	**0.659 (0.029)**	**0.402 (0.013)**	**0.686 (0.008)**	**0.893 (0.007)**	**0.660**	**0.402**	**0.893**

4.6 dcBiLSTM-AE embeddings correlation analysis

To assess representational fidelity, we compute four physicochemical properties on the dataset **S** test split (see **Section 3.2**). Targets were chosen to be analytically derived for interpretation clarity, yet being functionally relevant: instability index (ISI), theoretical net charge (TNC), isoelectric point (IEP), and molecular weight (MW). Correlating these clear whole-peptide properties with interpretable feature space reveals whether embeddings align with monomer-level descriptors or not (**Table 3**). The same protocol was applied to earlier encodings (**Table 14**).

5 DISCUSSION

5.1 MODEL SCREENING AND OPTIMIZATION

Initial benchmarking of the most common peptide encoding strategies as well as the baseline CAE model on four benchmark datasets (**Table 1**) was performed to both identify the complexity of tasks and performance of the CAE model. According to the results, AMP and AOP (MCC = 0.499 and MCC = 0.503, respectively) are on average more successfully identified by the models **(Table 6)**, whereas ADP and AIP (MCC = 0.165 and MCC = 0.267) were more difficult, which can be due high Shannon entropies of sequences and Levenshtein distances between them given the number of samples insufficient to describe high compositional complexity. Baseline CAE demonstrated moderate performance, as presented in **Table 6**. Specifically, the CAE ranked fourth out of five in two tasks (AMP and AOP), third in one task (ADP), and second in the remaining task (AIP). Interestingly, the CAE model was ranked higher in more difficult tasks, competing with ProtBERT.

To enhance the quality of embeddings, we conducted experiments utilizing architectures for sequential data processing, namely BiLSTM and transformers. To alleviate potential latent space sparsity issues, we evaluated InfoCVAE model (Zhao et al., 2019) designed to mitigate inaccurate amortized inference and the tendency of VAEs to ignore latent variables when using flexible decoders. The results of the AE-based models screening stage are summarized in **Table 7**. CAE and BiLSTM-AE achieved comparable performance, with the transformer-based AE exhibiting less stable results; therefore, BiLSTM-AE model was selected for further investigation.

Motivated by the superior performance of one-hot encoded representations, we integrated contrastive learning into the training process (details in **Section 3.6**). As demonstrated in **Table 2**, this step led to a substantial improvement over both one-hot encoded representations and BiLSTM-AE.

To mitigate feature entanglement, we implemented a feature disentanglement procedure (**Section 3.7**). We initially hypothesized that the forced diagonal initialization of weights, combined with a penalty to preserve this structure, might lead to a reduction in predictive performance due to the constrained model capacity. Contrary to this expectation, the results in **Table 2** demonstrate that this regularization not only preserved but slightly improved the model's performance. This suggests that the imposed structural bias acts as a useful regularizer, effectively guiding the learning process. Furthermore, this approach provided a foundation for interpretability at the latent space level, as evidenced by a diagonality metric of 49.4% for the dcBiLSTM-AE model (**Section 3.7**).

These results demonstrate that our model consistently delivers stable and robust performance across the evaluated cases including ADPs and AIPs being the most problematic for all encoding strategies under the study, while showing latent space-level interpretability. For instance, although dominance in raw performance is not the main aim of this research, in two out of four tasks our method outperforms the ProtBERT model. This result is particularly remarkable given that our model has no preliminary fine-tuning step (thereby, excluding a trivial explanation that the model performs well due to overfitting) and operates with $>$4,000 times fewer trainable parameters. This drastic reduction in model complexity underscores the efficiency, while still achieving competitive and reliable performance across diverse tasks.

5.2 EMBEDDING SPACE INTERPRETABILITY STUDIES

To evaluate the interpretability of the obtained peptide embeddings, we conducted a regression analysis aimed at predicting a set of basic peptide properties (see **Section 4.5**). Additionally, we assessed the relationships between individual embedding elements and several biologically significant physicochemical properties of peptides using Pearson's correlation tests. The embeddings generated by our model were used as descriptors for an extreme gradient boosting regression model, which predicted the target properties based on these embeddings. The regression analysis demonstrated that the embeddings effectively captured relevant information about the peptide properties, where the results for the final dcBiLSTM-AE model were comparable with the other approaches (**Table 5**). Detailed results of this analysis are provided in **Table 2** of **Appendix C**. To further explore the interpretability of the embeddings, we examined the correlations between individual latent space features as well as the physicochemical properties. For each property, we identified the top 10 embedding elements with the highest absolute Pearson correlation coefficients (PCC). Among these, we highlighted the features with the most statistically significant correlations based on their p-values.

Overall, the analysis revealed that certain features consistently demonstrated strong correlations with specific physicochemical properties (**Table 3**; for a complete list see **Appendix D**), which can be interpreted as indirect yet a strong sign that the latent space features provide a substantial extent of interpretability. For instance, results for ISI show the strongest correlation of number of aromatic rings, saturated heterocycles and spiro atoms with the target value. Each of these parameters characterize connectivity within the molecules, therefore being related to branching, which directly influences peptide conformational flexibility and, therefore, the stability. Also, in case of TNC and IEP directly related to peptide charge, model shows that partition coefficient logP and Ca^{2+} interaction energy are highly relevant, where the former is a parameter describing protein solubility and the latter is strongly related to the presence and alignment of negatively charged amino acids in the protein. It is important to note, however, that not all the features have a clear and well-studied relationship with the properties predicted in these tasks, while for some of the tasks (e.g., MW), most of the parameters are directly related to the target value, and the model was unable to differentiate between them. Overall, these findings suggest that our model is able to capture meaningful features related to peptide properties, providing a foundation for interpreting its outputs in a biologically informed manner.

Table 3: Correlation between embedding dimensions and peptide properties.

Peptide property	Encoded input feature	PCC (r)
ISI	Num Arom Rings	-0.253
	Num Saturated Heterocycles	0.202
	Num Spiro Atoms	-0.193
TNC	CrippenClogP	-0.738
	Ca^{2+} interaction energy	0.542
	Num Atom Stereo Centers	-0.537
IEP	CrippenClogP	-0.620
	Ca^{2+} interaction energy	0.494
	Num Atom Stereo Centers	-0.392
MW	chi1v	-0.994
	chi2v	0.992
	kappa3	0.989

6 CONCLUSION

In this work, we propose a novel approach towards domain-tailored development of 3-4 orders more parameter-efficient pLMs trained on much smaller yet representative peptide datasets combining cross-task stability requiring no per-task fine-tuning and feature space-level interpretability for peptide representation learning. Although model scaling requires further research to compete with or surpass current SOTA ProtT5 model with input-level interpretability in raw performance, these findings hold great potential by introducing explainable latent spaces crucial for domain scientists building their own predictive models on specific downstream tasks, which is not provided by any of the pLM models existing to date, at no performance cost.

REPRODUCIBILITY STATEMENT

The code and datasets used in this study are available to ensure reproducibility of the presented results. They can be accessed at the following repository: `https://anonymous.4open.science/r/SeQuant-perfomance-1822/`.

REFERENCES

Shaherin Basith, Balachandran Manavalan, Tae Hwan Shin, and Gwang Lee. Machine intelligence in peptide therapeutics: A next-generation tool for rapid disease screening. *Medicinal research reviews*, 40(4):1276–1314, 2020.

Qiushi Cao, Cheng Ge, Xuejie Wang, Peta J Harvey, Zixuan Zhang, Yuan Ma, Xianghong Wang, Xinying Jia, Mehdi Mobli, David J Craik, et al. Designing antimicrobial peptides using deep learning and molecular dynamic simulations. *Briefings in Bioinformatics*, 24(2):bbad058, 2023.

Xue Chen, Jian Huang, and Bifang He. Antidmppred: a web service for identifying anti-diabetic peptides. *PeerJ*, 10:e13581, 2022.

Davide Chicco and Giuseppe Jurman. The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation. *BMC genomics*, 21(1):6, 2020.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171–4186, 2019.

Claudia Draxl and Matthias Scheffler. The nomad laboratory: from data sharing to artificial intelligence. *Journal of Physics: Materials*, 2(3):036001, 2019.

Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. Prottrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381.

Ahmed Elnaggar, Hazem Essam, Wafaa Salah-Eldin, Walid Moustafa, Mohamed Elkerdawy, Charlotte Rochereau, and Burkhard Rost. Ankh: Optimized protein language model unlocks general-purpose modelling. *arXiv preprint arXiv:2301.06568*, 2023.

ESM Team. Esm cambrian: Revealing the mysteries of proteins with unsupervised learning, 2024. URL `https://evolutionaryscale.ai/blog/esm-cambrian`.

Elizaveta Fisher, Kirill Pavlenko, Alexander Vlasov, and Galina Ramenskaya. Peptide-based therapeutics for oncology. *Pharmaceutical medicine*, 33(1):9–20, 2019.

Jure Gregorc, Nathalie Lensen, Grégory Chaume, Jernej Iskra, and Thierry Brigaud. Trifluoromethylthiolation of tryptophan and tyrosine derivatives: A tool for enhancing the local hydrophobicity of peptides. *The Journal of Organic Chemistry*, 88(18):13169–13177, 2023.

Antoine Henninot, James C Collins, and John M Nuss. The current state of peptide drug discovery: back to the future? *Journal of medicinal chemistry*, 61(4):1382–1414, 2018.

Xiaojuan Hu, Maja-Olivia Lenz-Himmer, and Carsten Baldauf. Better force fields start with better data: A data set of cation dipeptide interactions. *Scientific Data*, 9(1):327, 2022.

Sarthak Jain and Byron C Wallace. Attention is not explanation. *arXiv preprint arXiv:1902.10186*, 2019.

Kjell Jorner, Anna Tomberg, Christoph Bauer, Christian Sköld, and Per-Ola Norrby. Organic reactivity from mechanism to machine learning. *Nature Reviews Chemistry*, 5(4):240–255, 2021.

G Landrum et al. Rdkit: cheminformatics and machine learning software. *RDKIT. ORG*, 232, 2013.

Byungju Lee, Jaekyun Yoo, and Kisuk Kang. Predicting the chemical reactivity of organic materials using a machine-learning approach. *Chemical science*, 11(30):7813–7822, 2020.

Hirotomo Moriwaki, Yu-Shi Tian, Norihito Kawashita, and Tatsuya Takagi. Mordred: a molecular descriptor calculator. *Journal of cheminformatics*, 10(1):4, 2018.

Shaylee C Peterson and Arden R Barry. Effect of glucagon-like peptide-1 receptor agonists on all-cause mortality and cardiovascular outcomes: a meta-analysis. *Current Diabetes Reviews*, 14 (3):273–279, 2018.

Dongya Qin, Linna Jiao, Ruihong Wang, Yi Zhao, Youjin Hao, and Guizhao Liang. Prediction of antioxidant peptides using a quantitative structure- activity relationship predictor (anoxpp) based on bidirectional long short-term memory neural network and interpretable amino acid descriptors. *Computers in Biology and Medicine*, 154:106591, 2023.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020.

Ali Raza, Jamal Uddin, Abdullah Almuhaimeed, Shahid Akbar, Quan Zou, and Ashfaq Ahmad. Aips-sntcn: Predicting anti-inflammatory peptides using fasttext and transformer encoder-based hybrid word embedding with self-normalized temporal convolutional networks. *Journal of chemical information and modeling*, 63(21):6537–6554, 2023.

Taha Rezai, Jonathan E Bock, Mai V Zhou, Chakrapani Kalyanaraman, R Scott Lokey, and Matthew P Jacobson. Conformational flexibility, internal hydrogen bonding, and passive membrane permeability: successful in silico prediction of the relative permeabilities of cyclic peptides. *Journal of the American Chemical Society*, 128(43):14073–14080, 2006.

Eric W Sayers, Jeff Beck, Evan E Bolton, J Rodney Brister, Jessica Chan, Donald C Comeau, Ryan Connor, Michael DiCuccio, Catherine M Farrell, Michael Feldgarden, et al. Database resources of the national center for biotechnology information. *Nucleic acids research*, 52(D1):D33, 2023.

Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022.

Marcelo DT Torres, Shanmugapriya Sothiselvam, Timothy K Lu, and Cesar de la Fuente-Nunez. Peptide design principles for antimicrobial applications. *Journal of molecular biology*, 431(18): 3547–3567, 2019.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

Clara G Victorio and Nicholas Sawyer. Folding-assisted peptide disulfide formation and dimerization. *ACS Chemical Biology*, 18(7):1480–1486, 2023.

David Weininger. Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of chemical information and computer sciences*, 28(1):31–36, 1988.

Yuting Xu, Deeptak Verma, Robert P Sheridan, Andy Liaw, Junshui Ma, Nicholas M Marshall, John McIntosh, Edward C Sherer, Vladimir Svetnik, and Jennifer M Johnston. Deep dive into machine learning models for protein engineering. *Journal of chemical information and modeling*, 60(6): 2773–2790, 2020.

Shengjia Zhao, Jiaming Song, and Stefano Ermon. Infovae: Balancing learning and inference in variational autoencoders. In *Proceedings of the aaai conference on artificial intelligence*, volume 33, pp. 5885–5892, 2019.